

MATB24H3

Exercise Questions

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1 Notation/Terminologies

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2 Fields & Rings & Groups

2.1 Preliminary

2.1.1. (pg. 6) Consider the set $X = \mathbb{Q} \setminus \{0\}$ and the map $\star$ is depicted by $a \star b = \frac{a}{b}$. Is $\star$ associative?

Solution:

2.1.2. (pg. 7) Is $(\mathcal{P}(\mathbb{N}), \cap)$ a monoid? What is the identity element? How about $(\mathcal{P}(X), \cup)$ for any arbitrary set X ?

Solution:

2.2 Fields

2.2.1 (pg. 13) Suppose d is square-free, so there is no integer m such that $m^2 = d$. Show that $(\mathbb{Q}(\sqrt{d}), +, \cdot)$ forms a field. What happens if $d = m^2$ for some m ?

Solution:

2.2.2. (pg. 12) Now, can we extend a *finite field* in the same way we extended an infinite field? If so, give an example.

Solution:

2.2.3. (pg. 14) Prove that $0_{\mathbb{F}}$, $1_{\mathbb{F}}$ and additive inverse element $-a$ of any element $a \in \mathbb{F}$ are unique.

Solution:

2.2.4. (pg. 15) Prove $(-a)(-b) = ab$ for all $a, b \in \mathbb{F}$.

Solution:

2.2.5. (pg. 16) Prove that the triple $(\mathbb{Z}_p, + \bmod p, \cdot \bmod p)$ is a field $\iff p$ is a prime. Use the fact that if two integers m, n are relatively prime, then there exist integers r and s such that $mr + ns = 1$.

Solution:

2.2.6. (pg. 20) Let,

$$S = \{r^0 := 1, r, r^2, r^3\}$$

Prove that the 2-tuple $(S, \circ)$, where S is the set of symmetries of a square and $\circ$ is the composition operation, does indeed form a group. Moreover, find the symmetry group of a rectangle.

Solution:

2.2.7. (pg. 22) Characterize all idempotent elements of $(\mathcal{M}_{2 \times 2}(\mathbb{R}), +, \cdot)$.

Solution:

2.2.8. (pg. 26) Prove that $\mathcal{C}(\mathcal{R})$ is a subring of any ring $(\mathcal{R}, +, \cdot)$.

Solution:

2.2.9. (pg. 26) Let $(\mathcal{R}, +, \cdot)$ be a ring and suppose that $\alpha \in \mathcal{R}$ has a multiplicative inverse. Moreover, suppose that $\alpha \in C(\mathcal{R})$. Is $\alpha^{-1} \in C(\mathcal{R})$?

Solution:

2.2.10. (pg. 28) Does the set of functions with infinitely many roots form a subring of $\mathcal{F}(\mathbb{R})$? Justify your answer.

Solution:

[Refer to notes] (pg. 29)

2.2.12. (pg. 29) Determine the zero and the identity elements of the polynomial ring $(\mathcal{R}[x], +, \cdot)$ where $\mathcal{R} = \mathcal{C}(\mathcal{M}_{2 \times 2}(\mathbb{R}))$.

Solution:

2.2.13. (pg. 32) True/False. Justify:

1. Let $\mathbb{F}$ be any field. Suppose that α and β are two elements of an extension field $\mathbb{K}$ of $\mathbb{F}$. Then $\mathbb{F}(\alpha)$ and $\mathbb{F}(\beta)$ are always distinct extension fields of $\mathbb{F}$.
2. Given any set X , there is no operation $\cdot$ that can be defined on $[X]^{\leq |X|}$ that makes it a group.
3. Given a monoid $(\mathcal{S}, *)$ there exists a binary operation $\cdot$ on $\mathcal{S}$ such that $(\mathcal{S}, \cdot)$ no longer admits an identity element. If so, what assumption did you need to make?
4. $(\mathbb{R}, \cdot)$ forms a group.
5. $(\mathbb{R}, +)$ forms a group with the identity element being 1.
6. Suppose $(\mathbb{F}, +, \cdot)$ is the field. Then $(\mathbb{F}, +)$ and $(\mathbb{F}, \cdot)$ both form a group.

1) *Solution:*

2) *Solution:*

3) *Solution:*

4) *Solution:*

5) *Solution:*

6) *Solution:*

3 Vector Spaces

3.1 Basis

3.1.1. (pg. 37) Had we defined vector addition by

$$\hat{A} \boxplus \hat{B} = [a_{ij} \cdot b_{ij}] \text{ i.e., pointwise multiplication of the entries}$$

in Example 3.1.3, would

$$(\mathcal{M}_{m \times n}(\mathbb{F}), \boxplus, \odot, (\mathbb{F}, +, \cdot))$$

form a vector space over $\mathbb{F}$?

Solution:

3.1.2. (pg. 40) Prove properties 2, 4 and 5 of Theorem 3.1.2, namely,

For any vector space $\mathbb{V} = (V, \boxplus, \odot, (\mathbb{F}, +, \cdot))$ over field $\mathbb{F}$:

- 2) The additive inverse of any $v \in V$ is unique.
- 4) Any scalar of the zero vector yields the zero vector, i.e. $\forall \alpha \in \mathbb{F}, \alpha \odot \vec{0} = \vec{0}$.
- 5) For all $v \in V$ and $c \in \mathbb{F}$, $(-c) \odot v = c \odot (-v) = -1 \odot (c \odot v)$.

Solution:

3.2 Subspaces

3.2.1. (pg. 42) Prove Theorem 3.2.2, namely

If $S \subseteq V$ then $\text{span}(S) \subseteq V$ is a subspace of V .

Solution:

3.2.2. (pg. 42) Prove Theorem 3.2.3, namely:

Let J be any index set (It is a set whose elements are used to enumerate the elements of set). Let $\{W_\alpha \mid \alpha \in J\}$ be a collection of subspaces of V . Then $\bigcap_{\alpha \in J} W_\alpha$ is a subspace of V .

Solution:

3.2.3. (pg. 43) Prove Theorem 3.2.4, namely:

Let $W_1 \subseteq W_2 \subseteq W_3 \subseteq \dots$ be an increasing chain of subspaces of V . Then $\bigcup_{n \in \mathbb{N}} W_n$ is a subspace.

Solution:

3.3 Bases

3.3.1. (pg. 46) Show that the set $\{e^x, e^{2x}, e^{3x}\}$ is linearly independent in the vector space of $\mathcal{F}(\mathbb{R})$ over $\mathbb{R}$.

Solution:

3.4 Linear Transformations

3.4.1. (pg. 50) Prove Proposition 3.4.1, namely,

The following: Let $T: V \rightarrow W$ be a linear transformation from V to W . Then:

1. For all $x, y \in V$, $T(x - y) = T(x) - T(y)$
2. T maps the zero vector $\vec{0}_V$ of V to the zero vector $\vec{0}_W$ of W .
3. If $T': W \rightarrow Z$ is also a linear transformation, their composition $T' \circ T$ is also a linear transformation from V to Z where $T' \circ T = T'(T(v))$.
4. If V_1 is a subspace of V and W_1 is a subspace of W , then $T[V_1]$ is a subspace of W and $T^{-1}[W_1]$ is a subspace of V .

1) *Solution:*

2) *Solution:*

3) *Solution:*

4) *Solution:*

3.4.2. (pg. 51) Prove Theorem 2.4.2

(currently, this theorem does not exist on the lecture notes, but Pourya will surely fix this in no time.)

Solution:

3.4.3. (pg. 52) Prove Theorem 3.4.2, namely,

show that $(\mathcal{L}(V, W), \boxplus, \odot, (\mathbb{F}, +, \cdot))$ forms a vector space over $\mathbb{F}$, where $\boxplus$ and $\odot$ are the pointwise addition of linear maps and scalar multiplication with the linear map in W , respectively. That being, $\boxplus$ is the vector addition operation in W and $\odot$ is the scalar multiplication operation in W .

Solution:

3.4.4. (pg. 53) Prove that a linear map $T: V \rightarrow W$ is invertible $\iff T$ is injective and surjective.

Solution:

3.4.5. (pg. 53) Let V and W be two vector spaces over a field $\mathbb{F}$ with dimensions n and $m > 3$ respectively. Construct three different linearly independent linear maps from V to W .

Solution:

3.4.6. (pg. 53) Prove whether or not the triple $(\mathcal{L}(\mathbb{R}, \mathbb{R}), +, \cdot)$ forms a subring of $\mathcal{F}(\mathbb{R})$ under point-wise addition and multiplication.

Solution: